

☒ Full Theory: Attacks, IDS/IPS, and Threat Hunting

☒ 1. Attacks: Traditional vs Distributed

☒ Traditional Attacks:

- Launched from a single system
- Easy to trace back to source
- Examples:
 - Port scanning
 - Password cracking
 -

Malware injection

☒ Distributed Attacks (DDoS):

- Multiple compromised systems (botnet) attack one target
- Harder to defend due to scale
- Examples:
 - Distributed Denial of Service (DDoS)
 - Coordinated brute force
 - Distributed scanning

❏ 2. Types of Intruders

Intruder Type	Description
Masquerader	Not authorized, but gains access using stolen credentials
Misfeasor	Legitimate user who abuses privileges
Clandestine user	Tries to avoid detection (e.g., backdoor access)
External attacker	No internal access; breaches perimeter
Insider attacker	Has authorized access, uses it maliciously

❏ 3. Types of IDS

IDS Type	Description
NIDS – Network IDS	Monitors packets on a network
HIDS – Host IDS	Monitors a host's logs, activities
WIDS – Wireless IDS	Detects rogue APs, wireless threats
Application IDS	Secures application layer activity
Hybrid IDS	Combines NIDS and HIDS for broad protection

❏ 4. IPS Categories

Category	Function
Network-based IPS (NIPS)	Deployed inline on network perimeter
Host-based IPS (HIPS)	Installed on end devices, blocks local threats
Wireless IPS (WIPS)	Prevents wireless intrusions
Network Behavior Analysis (NBA)	Detects anomalies in traffic behavior
Content-Based IPS	Filters based on payload (e.g., antivirus signatures)

☒ 5. Defense in Depth (DiD)

- Multi-layered security approach
- If one control fails, another protects the system

Layers include:

- Physical
- Network (firewalls, IPS)
-

Endpoint (antivirus, HIDS)

- Application (WAF, secure coding)
- Data (encryption)
- User (training, awareness)

☒ *Goal: no single point of failure.*

☒ 6. IDS and IPS Analysis Scheme

Key stages:

1. **Data Collection** – Packet sniffing, system logs
- 2.

Preprocessing – Filtering noise, normalizing data

3.

Detection Engine – Matches data with attack signatures or behavior anomalies

4.

Alerting/Action – Logs event, sends alert, or blocks traffic

5.

Post-Analysis – Correlation and investigation

☒ 7. Detection Methodologies

Method	Description	Pros	Cons
Signature-based	Detects known attacks using patterns	Fast, accurate	Misses unknown (zero-day)
Anomaly-based	Learns normal behavior, flags deviations	Detects unknown	High false positives
Heuristic/Behavioral	Uses rules/AI to identify suspicious behavior	Adaptive	Complex, slower
Hybrid	Combines multiple methods	Best coverage	Higher resource use

☒ 8. Principles of IDS

1.

Transparency – Should not interfere with network operations

2.

Timeliness – Quick detection and alerting

3.

Accuracy – Balance between false positives and false negatives

4.

Scalability – Must work in large networks

5.

Resilience – Must withstand evasion tactics and DoS

6.

Stealth – Must avoid detection by intruders

☒ 9. Threat Hunting Model

- **Proactive search** for threats that evade traditional tools
- Based on **hypotheses, telemetry data, and attacker TTPs**

Stages:

1. **Trigger** – Alert, anomaly, or hypothesis starts the hunt
2. **Investigation** – Querying logs, endpoint data
3. **Pattern Discovery** – Looking for indicators of compromise (IOCs)
- 4.

Enrichment – Adding threat intel context

5.

Remediation – Blocking, patching, or isolating assets

Tools used:

SIEM, EDR, Threat Intelligence Feeds, MITRE ATT&CK Framework